

Fine Tuning Assignments

Assignment 1 – Full Fine Tuning and deployment with Flask

Task

Build on the gpt2 fine tuning example provided for Ayurveda dataset.

Fine tune GPT2 model with full fine tuning

Deploy it using Flask and test it locally.

What to submit?

Screenshot of your python output for REST and openai API (python codes 4 and 5)

Assignment 2 – Instruction Fine Tuning with PEFT with qLoRA

Task

Use qLoRa to fine tune a base model to a specific instruction dataset. Establish a baseline and evaluate the model after fine tuning. Observe if the accuracy of the model improves after fine tuning or gets worse. Provide your analysis of the observation.

Which base model to use?

Any model which can be trained on free GPU resources on colab.

Tip: Use AI chatbot to find out which models will fit the bill

Which training dataset to use?

Use the dataset based on the first letter of your name:

See the dataset_allocation excel spreadsheet

Implementation tips

Reference notebook is provided for the above.

It was created with the help of Claude 4 using following prompt:

Create a python notebook that demonstrates peft fine tuning lora technique using Hugging Face's PEFT library. Use any dataset from huggingface such as <https://huggingface.co/datasets/knkarthick/dialogsum> or similar. The training should happen on colab free resources - so choose a smaller base model, and qLORA technique to train. Use golden dataset and demonstrate evaluation metrics such as bertscore before and after the fine tuning.

What to submit?

Colab notebook demonstrating the above in the shared drive folder under “finetuning”

Assignment 3 – Alignment Fine Tuning with Reward model & PPO

Task

Fine tune a model with human/AI preferred responses so that the model produces preferred outputs (such as “movie is great” instead of “movie is good”. You will use Proximal Policy Optimization (PPO) technique to train the base model

You will need weights and biases API Key

<https://wandb.ai/site>

Which base model to use?

Any model which can be trained on free GPU resources on colab.

Tip: Use AI chatbot to find out which models will fit the bill

Which training dataset to use?

Use the dataset based on the first letter of your name:

See the PPO_dataset_allocation excel spreadsheet

Implementation tips

1. Train a tiny **reward model** (e.g., 2-layer MLP on frozen ϕ embeddings) on 2–5 k labelled examples.
2. Hand the reward fn to `trl.PPOTrainer`.

A 3-epoch PPO loop on ~50 k tokens already shows an upward reward curve you can plot live.

Reference notebook is provided.

What to submit?

Colab notebook demonstrating the above in the shared drive folder under “finetuning”

Assignment 4 – Alignment Fine Tuning with DPO

Task

Fine tune a model with human/AI preferred responses so that the model produces preferred outputs (such as “movie is great” instead of “movie is good”).

Which base model to use?

Any model which can be trained on free GPU resources on colab.

Tip: Use AI chatbot to find out which models will fit the bill

Which training dataset to use?

Use the dataset based on the first letter of your name:

See the DPO_dataset_allocation excel spreadsheet

Implementation tips

DPO training is a drop-in replacement for SFT—just feed (prompt, chosen, rejected) triples to `trl.DPOTrainer`, skip the reward model, and evaluate with win-rate or task metric.

Reference notebook is provided.

What to submit?

Colab notebook demonstrating the above in the shared drive folder under “finetuning”